

Stateless vs Stateful Processing in Spark Structured Streaming

This document explains whether the given Spark Structured Streaming pipeline is stateless or stateful, and why.

Short Answer

The pipeline is stateful. Even without aggregation, windowing, or groupBy, a stream-stream join makes the pipeline stateful.

Step-by-Step Classification

1. Order Stream

Streaming source only. No aggregation. No join. Stateless.

2. Status Stream

Streaming source only. No aggregation. No join. Stateless.

3. User DataFrame (Static)

Static batch data. No buffering required. Stateless.

4. Stream-Stream INNER JOIN

This operation is stateful. Spark buffers rows from both streams until matches arrive.

```
order_stream_df.join(status_stream_df, "order_id", "inner")
```

5. Stream-Static JOIN

This join is stateless because the static side is finite and immediately available.

Final Classification

Pipeline Part	Stateless	Stateful
Stream sources	Yes	No
Stream-Stream INNER join	No	Yes
Stream-Static join	Yes	No
Entire pipeline	No	Yes

Why No Watermark Is Required

The join type is INNER. Spark can safely drop unmatched rows and keep state bounded. Therefore, no watermark is required, but the operation remains stateful.

Key Rule to Remember

Stateful does not mean aggregation only. Stream-stream joins are also stateful.

One-Line Verdict

This pipeline is stateful because it contains a stream-stream join, even without aggregation or windowing.